In [1]:
```
import pandas as pd
```

In [4]:
```
pd.__version__
```

Out[4]:
```
'2.3.3'
```

In [14]:
```
emp = pd.read_excel(r"C:\Users\srini\Downloads\Rawdata.xlsx")
```

In [15]:
```
emp
```

Out[15]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^$0	10+

In [17]:
```
id(emp)
```

Out[17]:
```
1982358431616
```

In [18]:
```
emp.head()
```

Out[18]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

In [19]:
```
emp.shape
```

Out[19]:
```
(6, 6)
```

In [20]:
```
emp.tail
```

Out[20]:
```
<bound method NDFrame.tail of      Name          Domain       Age   Location   Sal
ary      Exp
0    Mike   Datascience#$  34 years     Mumbai   5^00#0       2+
1  Teddy^         Testing    45' yr  Bangalore  10%%000       <3
2   Uma#r  Dataanalyst^^#       NaN        NaN  1$5%000   4> yrs
3    Jane     Ana^^lytics       NaN   Hyderbad   2000^0      NaN
4  Uttam*      Statistics     67-yr        NaN   30000-  5+ year
5     Kim             NLP      55yr      Delhi  6000^$0      10+>
```

In [21]:
```
emp.isnull()
```

Out[21]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [22]:
```
emp.isna()
```

Out[22]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [23]:
```
emp.isnull().sum()
```

Out[23]:
```
Name        0
Domain      0
Age         2
Location    2
Salary      0
Exp         1
dtype: int64
```

In [24]:
```
emp.dtypes
```

Out[24]:
```
Name        object
Domain      object
Age         object
Location    object
Salary      object
Exp         object
dtype: object
```

In [25]:
```
emp['Name']
```

Out[25]:
```
0      Mike
1    Teddy^
2     Uma#r
3      Jane
4    Uttam*
5       Kim
Name: Name, dtype: object
```

In [26]:
```
emp['Name']= emp['Name'].str.replace(r'\W','',regex=True)
```

In [27]:
```
emp['Name']
```

Out[27]:
```
0     Mike
1    Teddy
2     Umar
3     Jane
4    Uttam
5      Kim
Name: Name, dtype: object
```

In [28]:
```
emp['Domain']= emp['Domain'].str.replace(r'\W','',regex=True)
```

In [29]:
```
emp['Domain']
```

Out[29]:
```
0    Datascience
1        Testing
2    Dataanalyst
3      Analytics
4     Statistics
5            NLP
Name: Domain, dtype: object
```

In [30]:
```
emp
```

Out[30]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^$0	10+

In [35]:
```
emp['Age'] = emp['Age'].str.replace(r'\W','',regex=True)
```

In [36]:
```
emp
```

Out[36]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34years	Mumbai	5^00#0	2+
1	Teddy	Testing	45yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^$0	10+

In [37]:
```
emp['Age'] = emp['Age'].str.replace(r'\W','',regex=True)
```

In [38]:
```
emp['Age']
```

Out[38]:
```
0    34years
1       45yr
2        NaN
3        NaN
4       67yr
5       55yr
Name: Age, dtype: object
```

In [ ]:
```
emp['Age'] = emp['Age'].str.extract(r'(\d+)')
```

In [ ]:
```
emp['Location']= emp['Location'].str.replace(r'\W','',regex=True)
```

In [39]:
```
emp
```

Out[39]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34years	Mumbai	5^00#0	2+
1	Teddy	Testing	45yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^$0	10+

In [40]:
```
emp['Salary'] = emp['Salary'].str.replace(r'\W','',regex=True)
```

In [41]:
```
emp['Salary']
```

Out[41]:
```
0     5000
1    10000
2    15000
3    20000
4    30000
5    60000
Name: Salary, dtype: object
```

In [42]:
```
emp['Exp']= emp['Exp'].str.extract(r'(\d+)')
```

In [43]:
```
emp
```

Out[43]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34years	Mumbai	5000	2
1	Teddy	Testing	45yr	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67yr	NaN	30000	5
5	Kim	NLP	55yr	Delhi	60000	10

In [44]:
```
clean_data = emp
```

In [45]:
```
clean_data
```

Out[45]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34years	Mumbai	5000	2
1	Teddy	Testing	45yr	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67yr	NaN	30000	5
5	Kim	NLP	55yr	Delhi	60000	10

In [ ]: